

YouTube Trending Video Analysis

Data Storytelling Report

Introduction

The rapid growth of digital video platforms has made it essential to understand what drives audience engagement and content popularity. This project focuses on analyzing YouTube trending video data to uncover patterns related to content categories, publishing behavior, sentiment, and regional performance. By combining data processing, analytical querying, and visualization techniques, the project aims to explain why certain videos trend and how engagement varies across genres and countries.

Abstract

This project analyzes over 160,000 YouTube trending videos using an end-to-end data analytics workflow. Python was used for data cleaning, transformation, and sentiment analysis on video titles and tags using the VADER sentiment analyzer. SQL was applied to aggregate key performance metrics such as average views by content category. Finally, Tableau dashboards were developed to visually explore trends, sentiment distribution, regional comparisons, and time-based performance. The analysis provides actionable insights into how content type, sentiment, and geography influence viewer engagement.

Tools Used:

- Python (Pandas, VADER Sentiment Analyzer):

Data cleaning, date standardization, feature engineering, and sentiment analysis on video titles and tags.

- SQL (MySQL):

Aggregation and analytical querying to calculate metrics such as average views by Category.

- Tableau:

Interactive dashboards for visual storytelling, trend analysis, and regional comparison.

Steps Involved in Building the Project:

1. Data Extraction and Cleaning:

Loaded YouTube trending data and verified data quality by checking schema, data types, and missing values. Standardized date fields and converted Boolean columns for database compatibility.

2. Sentiment Analysis:

Applied VADER sentiment analysis to classify titles and tags as positive, neutral, or negative, enabling sentiment-based engagement analysis.

3. SQL Analysis:

Aggregated data using SQL queries to calculate average views by content category and identify high-performing genres.

4. Data Visualization:

Designed Tableau dashboards to compare genres and sentiments, analyze regional engagement across countries, and observe time-series trends in average views.

Conclusion:

The analysis reveals that certain content categories consistently achieve higher average views, highlighting strong audience preferences for specific genres. Sentiment analysis shows that neutral and positive sentiments dominate trending content, while regional comparisons demonstrate significant variation in engagement levels across countries. Time-series analysis further indicates growth patterns in average views over time. These insights can help content creators, marketers, and analysts optimize publishing strategies, tailor content to audience preferences, and improve engagement outcomes. Future enhancements could include deeper category mapping and predictive modeling to forecast trending behavior.